

Airflow

Masterclass



Airflow



- What Airflow IS?
- What Airflow is NOT?
- When NOT to Use Airflow?

Building Blocks of Airflow

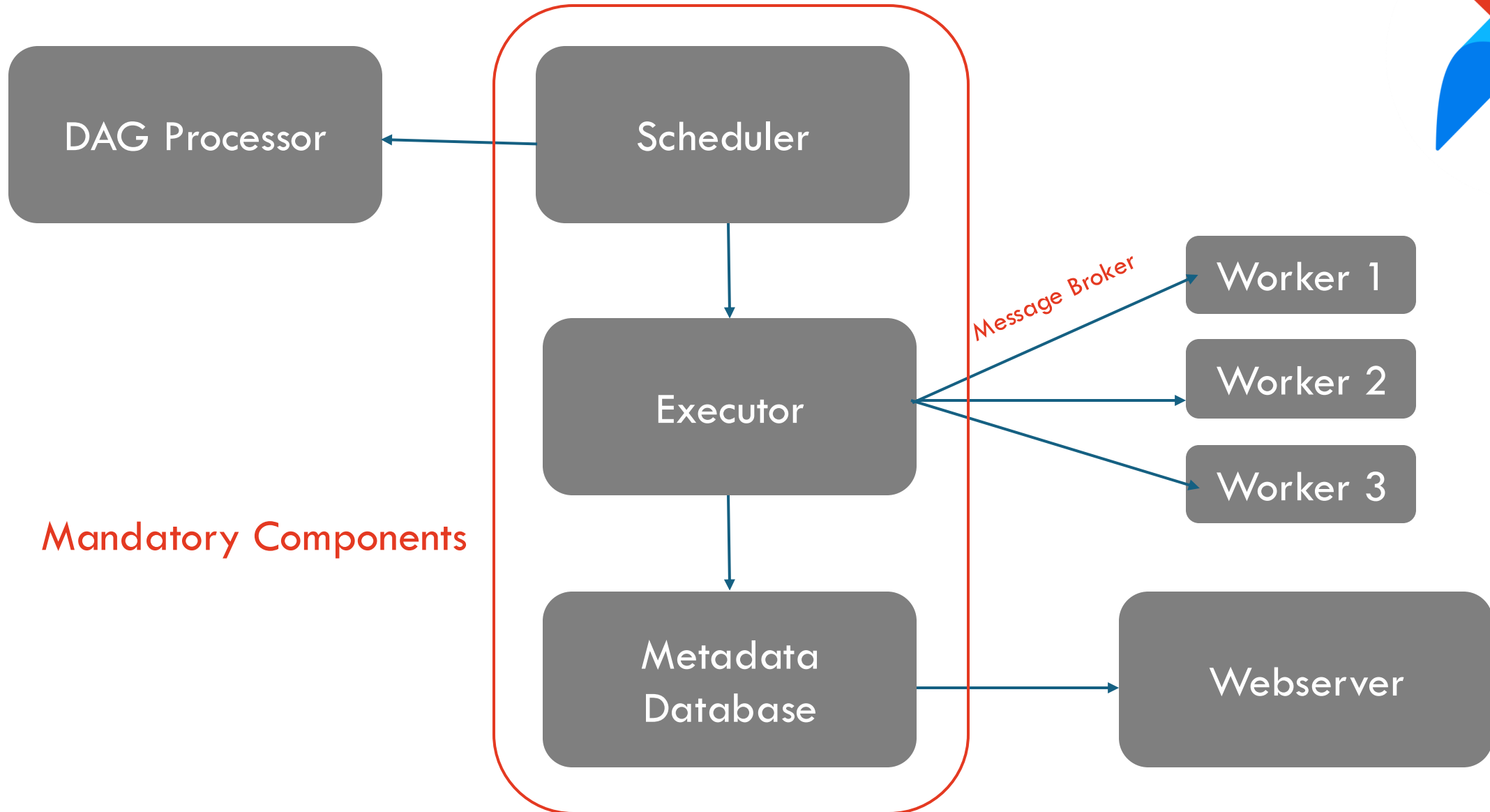


- DAG
 - Defines what should run and in what order
 - Python object
 - Static definition
- Task
 - A single unit of work
 - Node in the DAG
- Operator
 - Defines how a task does work
 - Python class (e.g., PythonOperator, BashOperator)

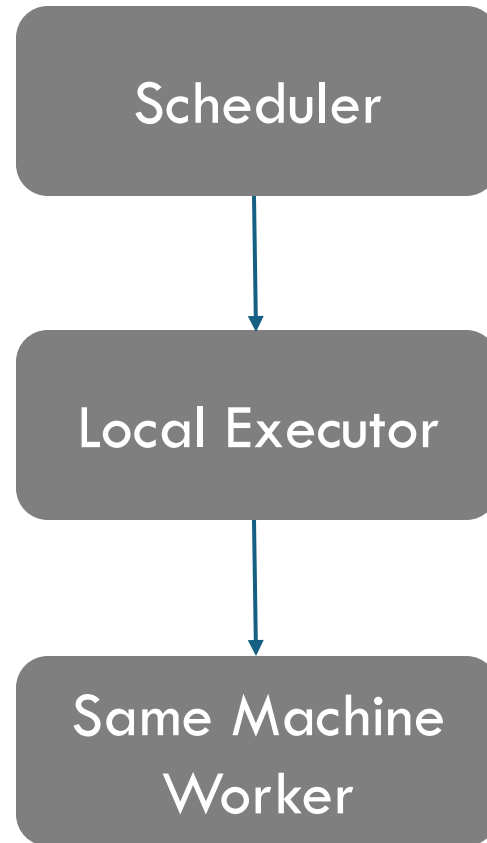
DAG Lifecycle



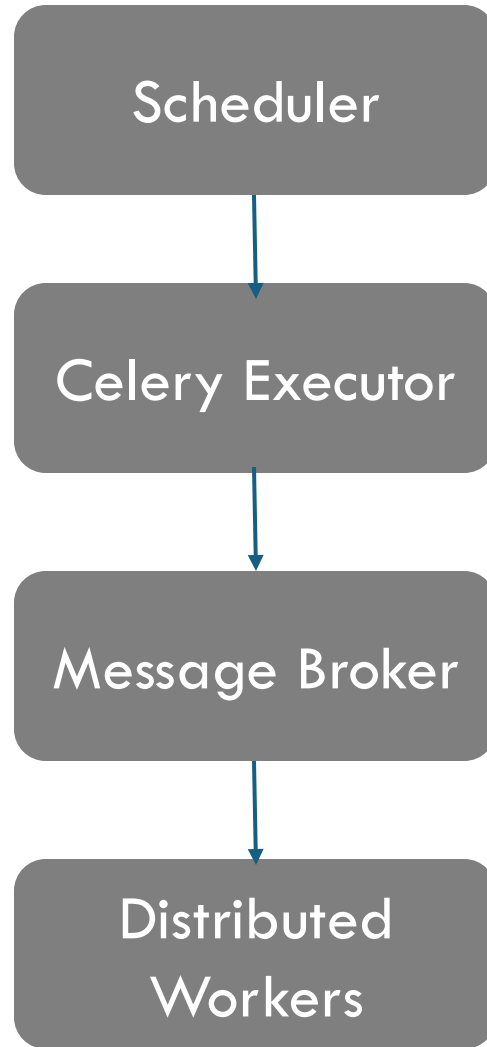
Airflow Architecture



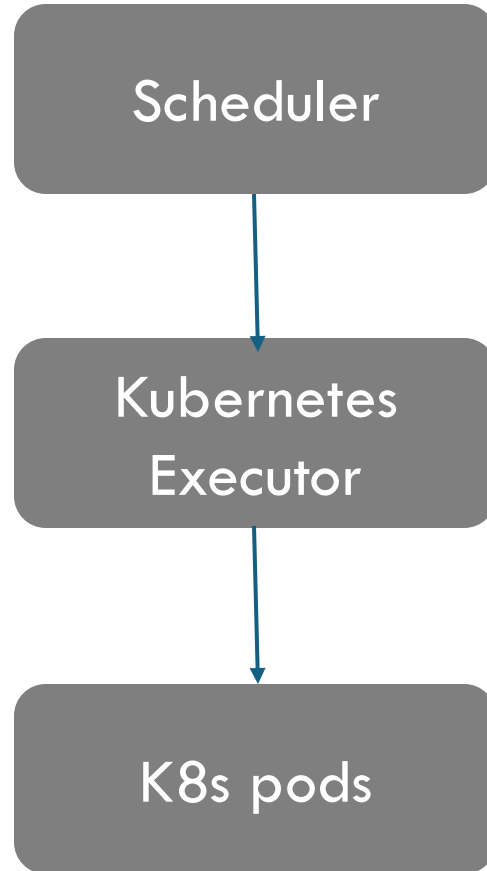
Local Executor



Celery Executor



Kubernetes Executor



Building Blocks of Airflow



Executor	How it Runs Tasks	Scalability	Setup Complexity	Typical Use
LocalExecutor	Same machine	Low	Very Low	Learning, POC
CeleryExecutor	Distributed workers	Medium–High	Medium	Classic production
KubernetesExecutor	One pod per task	Very High	High	Modern, cloud-native

Operator Categories



- Action Operators
 - Do something
 - Run code or command
- Transfer Operators
 - Move data between systems
- Sensor Operators
 - Wait for something to happen

XCom, Variables & Connections



- XCom
 - lets tasks exchange small pieces of runtime data during a DAG run.
- Variables
 - Key-value pairs stored in Airflow that externalize configuration so DAGs don't hard-code environment-specific values.
- Connections
 - Secure, reusable definitions for external systems that Airflow tasks use to authenticate and connect without exposing credentials.

Cron vs Timetables/ Catchup & Backfill



- **Cron** = defines when Airflow should create scheduled DAG runs.
- **Timetable** = explicitly controls how and when DAG runs are created.
- **Catchup** = Automatically creating missed historical DAG runs from the start_date up to the present.
- **Backfill** = Manually triggering DAG runs for past dates, independent of the scheduler.

How scheduling syntax works



- Preset schedules

- `schedule="@once"` # run once
- `schedule="@hourly"` # every hour
- `schedule="@daily"` # every day
- `schedule="@weekly"` # every week
- `schedule="@monthly"` # every month

- Cron expressions

0	1	2	3	4
minute	hour	day_of_month	month	day_of_week

*= Every

- At exact time.
- Multiple Specified Time.
- Range.
- Every n unit.

Field	Allowed values
minute	0–59
hour	0–23
day of month	1–31
month	1–12
day of week	0–6 (Sun–Sat)